

A reversible key uses each output letter once. One turn replaces every letter using that key; message positions stay fixed. Repeat the same key on each new message. Its return time is the first turn when all letters are back where they started.

Problem 1: Use letters A–H. Make a key for each return time: 4, 6, 8, 12, and 15 turns.

Problem 2: Make the slowest-returning keys you can with eight letters and with nine letters. Prove neither record can be beaten.

Problem 3: Does one extra letter always let you make a slower key? Find the greatest possible return time for each size from one to nine. Can adding a letter ever make this record smaller?